

HERMES — Complete Project Description

SIH26054 | AI-Enabled Real-Time Digital Twin for Health Monitoring, Fault Prediction and Mission Reliability of Aero-Piston Engines in MALE UAVs

1. Project Information

Project Name: Hermes

Problem Statement: SIH26054

Theme: Robotics and Drones

Focus: AI-enabled real-time digital twin for aero-piston engine health monitoring, fault prediction and mission reliability in MALE UAVs.

2. What is Hermes?

Hermes is a software-based system that monitors the health of an UAV engine during a mission.

It combines a physics-based engine model with AI and live telemetry.

It checks whether the real engine is behaving as expected.

It detects abnormal behaviour, identifies possible faults, estimates Remaining Useful Life (RUL), and tests what-if mission conditions.

The final decision remains with the human operator.

3. The Problem

Traditional engine monitoring mainly uses fixed thresholds.

Engine behaviour changes with altitude, load, RPM and mission phase.

A fixed limit may therefore create false alarms or miss early degradation.

Real engine fault data is also difficult to obtain because it is costly, limited and often proprietary.

Black-box AI can also be difficult for engineers to trust because it may not clearly explain why a fault was detected.

4. Hermes Solution

Hermes combines three main parts:

- Physics-based digital twin
- AI-based detection and prediction
- Mission-aware decision support

The system compares telemetry with expected engine behaviour and tracks the difference over time.

5. Digital Twin

A digital twin is a software model that represents the behaviour of a real engine.

Hermes uses the twin as a reference for normal engine behaviour.

Telemetry from the engine is compared with the model.

The difference between expected and actual behaviour is called a residual.

Growing residuals can indicate degradation or a fault.

6. Physics-Based Engine Model

The engine model represents important engine behaviour using physical relationships.

Hermes uses ODE-based modelling and a Mean Value Engine Model (MVEM).

This gives the AI a physical reference instead of relying only on historical data.

7. PINN and ODE

PINN means Physics-Informed Neural Network.

It combines neural-network learning with physical rules.

ODEs describe how engine states change with time.

The physics constraints help keep predictions connected to expected engine behaviour.

8. UKF

UKF means Unscented Kalman Filter.

It helps estimate engine states from noisy sensor measurements.

It combines the model's estimate with incoming telemetry.

This gives a more stable estimate of the engine's current condition.

9. Residual Tracking

Hermes calculates the difference between measured behaviour and model-predicted behaviour.

Small temporary differences may be normal.

A persistent or growing difference can indicate degradation.

Residuals are therefore used as an important health signal.

10. Degradation Detection

Hermes does not depend only on one unusual sensor value.

It considers persistence, trend and normalized residual size.

This helps separate short sensor disturbances from repeated abnormal behaviour.

The goal is to reduce unnecessary false alarms.

11. Anomaly Detection

Hermes uses physics-informed anomaly detection together with AI.

Isolation Forest can identify unusual patterns.

PINN-based analysis keeps the detection connected to physical engine behaviour.

This helps provide a more useful explanation than a simple abnormal/not-abnormal output.

12. Fault Classification

XGBoost is used for fault classification.

It can map detected patterns to known fault types.

The system can use multiple engine signals instead of depending on one sensor.

13. Unknown Fault Detection

Not every possible fault will be present in the training data.

Isolation Forest can help identify patterns that do not match known conditions.

These cases can be sent for further engineering investigation instead of being forced into a known fault class.

14. RUL Prediction

RUL means Remaining Useful Life.

Hermes estimates how much useful operating life may remain based on the current degradation state.

The estimate can change when the engine condition or mission scenario changes.

RUL is intended to support maintenance planning and mission decisions.

15. Mission-Aware Monitoring

Engine health is not independent of the mission.

Altitude, load, RPM and mission phase can change engine stress.

Hermes therefore considers the planned mission while evaluating engine risk.

This connects engine health to the actual operating situation.

16. Mission Sandbox

The Mission Sandbox allows an engineer or operator to test a what-if scenario.

For example, the system can compare the current mission profile with an alternative profile.

Hermes uses the current estimated engine condition to predict how the different scenario may affect engine health and RUL.

This allows decisions to be considered before the mission is carried out.

17. What-If Analysis

The sandbox is a counterfactual tool.

It asks: what could happen if a mission parameter or profile were changed?

The user can compare predicted outcomes rather than relying only on the current alarm.

18. Operator View

The operator sees the current engine condition and important alerts.

When Hermes detects an issue, the operator can see the recommended decision-support information.

The operator remains responsible for accepting or rejecting a recommendation.

19. Engineer View

The engineer view gives more detail about the detected issue.

It can show sensor trends, residuals, possible fault information and RUL.

The engineer can also use the Mission Sandbox to test alternative scenarios.

20. Maintenance View

Maintenance personnel can see degradation trends, RUL information and fault history.

This can support condition-based maintenance instead of relying only on fixed schedules.

The system can help identify which engine needs attention and why.

21. Complete Workflow

1. Telemetry enters Hermes.
2. The digital twin estimates expected engine behaviour.
3. Actual and expected behaviour are compared.
4. Persistent abnormal behaviour is detected.
5. AI and physics-based analysis help identify the fault.
6. RUL is estimated.
7. The Mission Sandbox tests possible mission scenarios.

8. The operator reviews the recommendation.
9. Maintenance receives the health and RUL information.

22. Prototype Data

The prototype can use synthetic and replayed telemetry.

Telemetry can be stored in CSV or Parquet format and replayed through MQTT.

This allows the system to be demonstrated without requiring access to a real UAV engine.

23. Why Synthetic Data?

Real engine fault data is difficult to collect.

Fault testing can be expensive and unsafe.

Many defence and aerospace datasets are proprietary.

Hermes can create physics-based fault-injected trajectories to produce labelled training and testing data.

24. Per-Engine Fingerprinting

Different engines can show slightly different normal behaviour.

Hermes can track the normal pattern of an individual engine.

This helps distinguish normal engine-to-engine variation from actual degradation.

25. Physics + AI

Physics alone can be difficult to model for every real-world condition.

AI alone may produce results that are difficult to explain.

Hermes combines both approaches.

Physics provides a reference and constraints, while AI helps detect patterns, classify faults and predict degradation.

26. Technology Stack

Backend: FastAPI and modular microservices.

Frontend: React-based interface.

Data: CSV/Parquet and time-series storage.

Telemetry: MQTT and replayed telemetry streams.

Models: PINN, ODE, MVEM, UKF, XGBoost, Isolation Forest and LSTM-based RUL prediction.

Deployment: Dockerized services.

27. Software Architecture

Hermes is designed as separate modules so components can be developed and deployed independently.

Telemetry handling, digital twin calculations, AI models, data storage and frontend services can be separated.

This makes the system easier to test and extend.

28. Hardware Approach

Hermes is software-first.

It aims to reuse existing UAV telemetry wherever possible.

Additional interface hardware is introduced only when required by the target UAV architecture.

This helps reduce added SWaP and integration cost.

29. GCS Integration

Hermes can be integrated with a Ground Control Station as an advisory system.

The system is intended to receive telemetry and provide health and mission information.

It does not need to directly control the aircraft.

30. Safety by Design

Hermes is designed as a decision-support system rather than an autonomous flight-control system.

It does not provide FADEC or ECU control feedback.

The human operator remains in control.

Any real deployment would require proper validation, certification, cybersecurity testing and approval.

31. Legal and Deployment Considerations

Real-world deployment must follow applicable aviation, defence, cybersecurity and data-security requirements.

Integration must respect the certified aircraft architecture.

Testing and validation are required before operational use.

32. Main Benefits

- Earlier detection of abnormal behaviour
- Better explanation of engine condition
- RUL-based maintenance support
- Mission-aware risk information
- What-if mission testing
- Reduced dependence on fixed thresholds
- Reuse of existing telemetry
- Scalable software architecture

33. Conventional Monitoring vs Hermes

Conventional systems mainly ask whether a parameter has crossed a limit.

Hermes also asks whether the engine is behaving differently from its expected physical behaviour.

Conventional alerts may stop at detection.

Hermes connects detection with diagnosis, RUL, mission simulation and maintenance information.

34. Main Innovation

The main idea is to connect engine health with mission decisions.

Hermes does not only say that an engine has a problem.

It tries to explain the problem, estimate its future effect and test alternative mission conditions.

35. Why the Mission Sandbox Matters

An engine may be degraded but still usable under some conditions.

The sandbox allows users to compare possible scenarios using the current engine state.

This turns engine monitoring into a planning and decision-support process.

36. Real-World Case

The project presentation includes a real-world MQ-1B Predator case from Djibouti in 2011.

The case is used to illustrate how repeated low-oil warnings and changing engine indications can be difficult to interpret using conventional monitoring.

Hermes uses this type of scenario to show the value of persistent trend analysis and health-aware decision support.

37. Impact

Hermes can support earlier maintenance action and better understanding of engine degradation.

It can help reduce unnecessary reactions to isolated sensor events.

It can connect maintenance information with mission planning.

For a fleet, the same software approach can be applied across multiple engines and aircraft, subject to proper model and data validation.

38. Project Feasibility

The prototype is software-focused and can run using synthetic or replayed telemetry.

Existing UAV telemetry can reduce the need for new onboard hardware.

Microservices make development modular.

The main challenge for real deployment is obtaining representative engine data and completing proper validation.

39. Main Risks

- Limited real fault data
- Difference between synthetic and real engine behaviour
- False alarms
- Incorrect RUL estimates
- Model generalization across engine types
- Integration with existing UAV systems
- Cybersecurity requirements
- Certification and validation requirements

40. Development Roadmap

Stage 1: Build the digital twin and prototype dashboard.

Stage 2: Add synthetic fault generation and AI models.

Stage 3: Validate using replayed and, where available, real telemetry.

Stage 4: Test mission sandbox scenarios.

Stage 5: Integrate with suitable GCS/test systems.

Stage 6: Complete detailed validation and deployment work.

41. Project Management

The project can be developed in separate modules.

Engine modelling, telemetry, AI, frontend, database and sandbox functions can be tested independently.

This allows the team to improve one part without rebuilding the entire system.

42. Business Model

Hermes can be offered as an engine-health and mission-intelligence software platform.

Possible revenue areas include software licensing, integration, customization, support and training.

The system does not need to be sold only as a replacement for an existing digital twin.

It can also work as a health and mission intelligence layer on top of existing systems.

43. Scalability

The same architecture can support different engine models by changing the physics model and relevant parameters.

The software can be expanded from one engine to a fleet.

New fault classes and new mission scenarios can be added over time.

44. Future Expansion

- More engine types
- More real telemetry
- Better RUL models
- More fault classes
- Fleet-level health monitoring
- More mission scenarios
- Improved GCS integrations
- Stronger validation with real test data

45. What Hermes Does NOT Claim

Hermes is not an autonomous flight-control system.

It does not directly control the engine or aircraft.

It does not replace certified pilots or maintenance engineers.

Prototype predictions should not be treated as operational certification results.

Real deployment requires proper testing and approval.

46. One-Minute Project Explanation

Hermes is an AI-enabled real-time digital twin for monitoring aero-piston engines used in MALE UAVs.

Traditional systems mainly use fixed thresholds, but engine behaviour changes with altitude, load, RPM and mission phase.

Hermes compares live telemetry with a physics-based digital twin and uses AI to detect abnormal behaviour, identify possible faults and estimate RUL.

It then uses a Mission Sandbox to test what-if mission scenarios using the current engine condition.

The operator can review the result and make the final decision, while maintenance teams receive degradation and RUL information.

In simple terms, Hermes detects the problem, explains it, predicts what may happen next, tests the future mission and supports maintenance planning.

47. Core USP

Hermes connects engine health with mission decisions.

Existing digital twins can show asset health.

Hermes aims to show what that health means for the mission and what could be done about it.

48. Final Project Summary

Hermes combines physics-based modelling, AI, telemetry and mission simulation into one decision-support system.

It is designed to detect problems early, understand why they may be happening, estimate remaining useful life, test possible mission scenarios, support the operator's decision and help maintenance teams plan action.

The overall goal is to move from simple threshold-based monitoring toward health-aware and mission-aware engine management.